

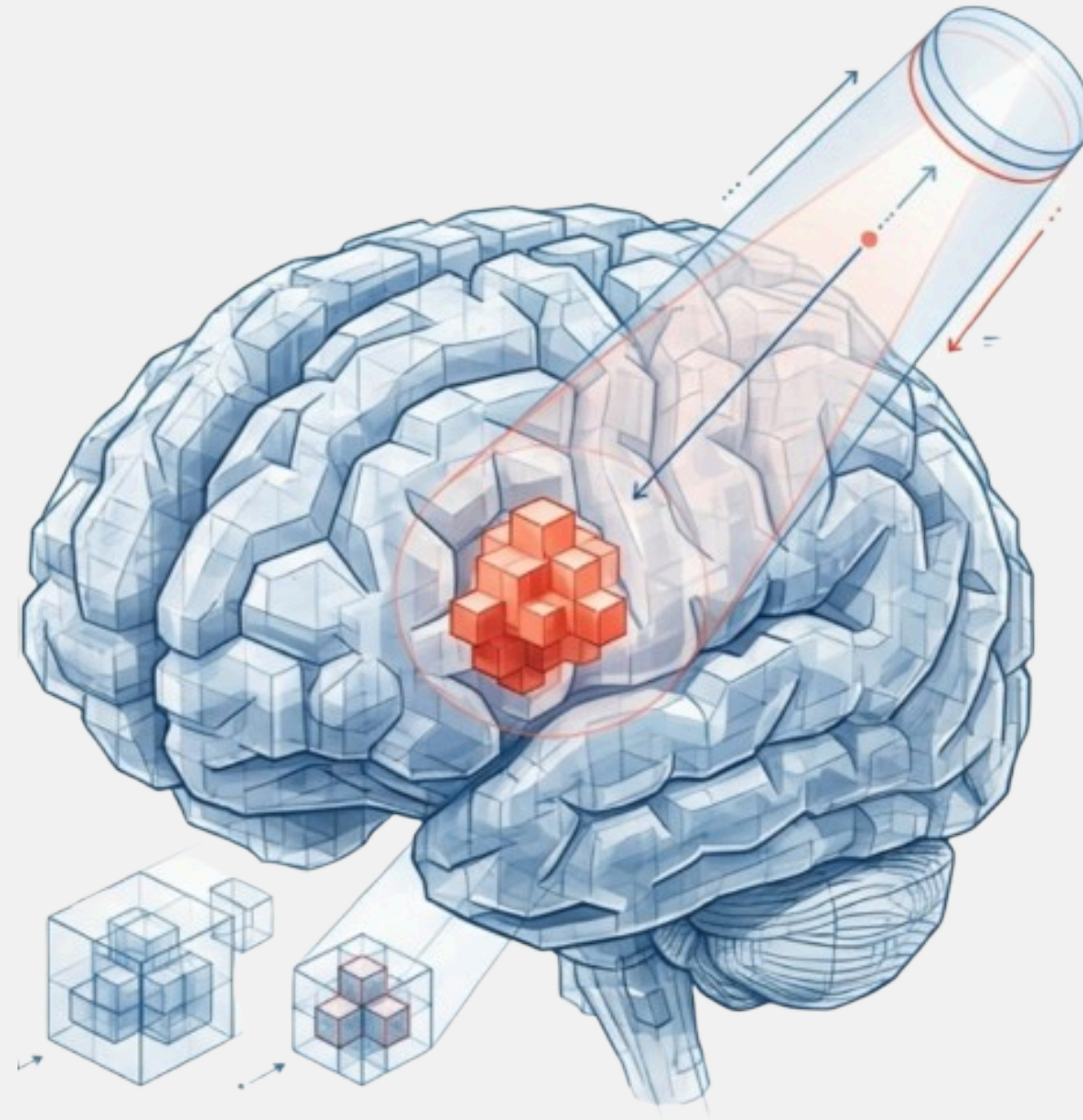


# Illuminating the Black Box

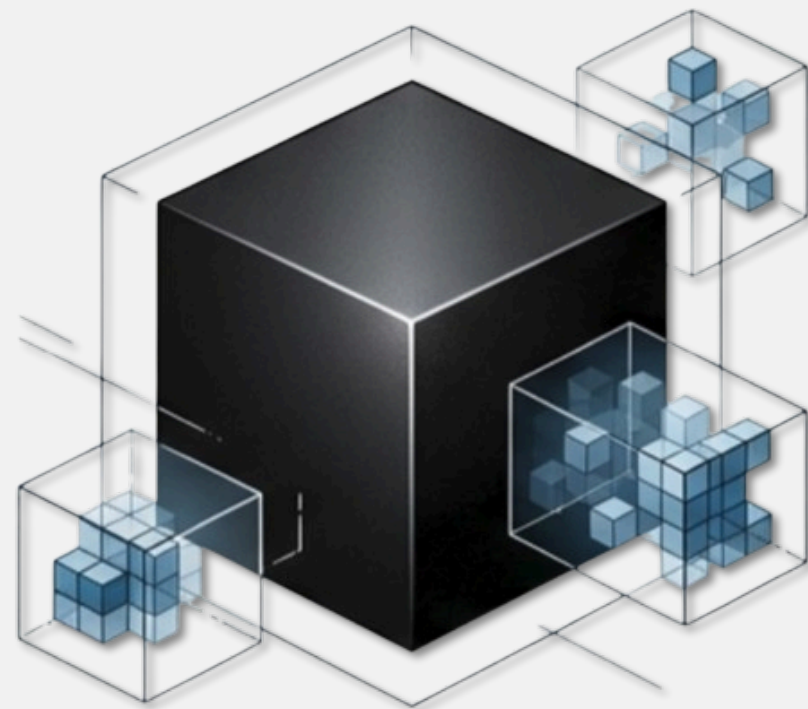
An Explainable Ai framework for brain tumor segmentation using volumetric data interpretation in 3D CNNs

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Supervisor Name: **Maimoona Maqsood SHARIF**



# The Trust Gap in High-stakes Neurosurgery



**The Black Box**

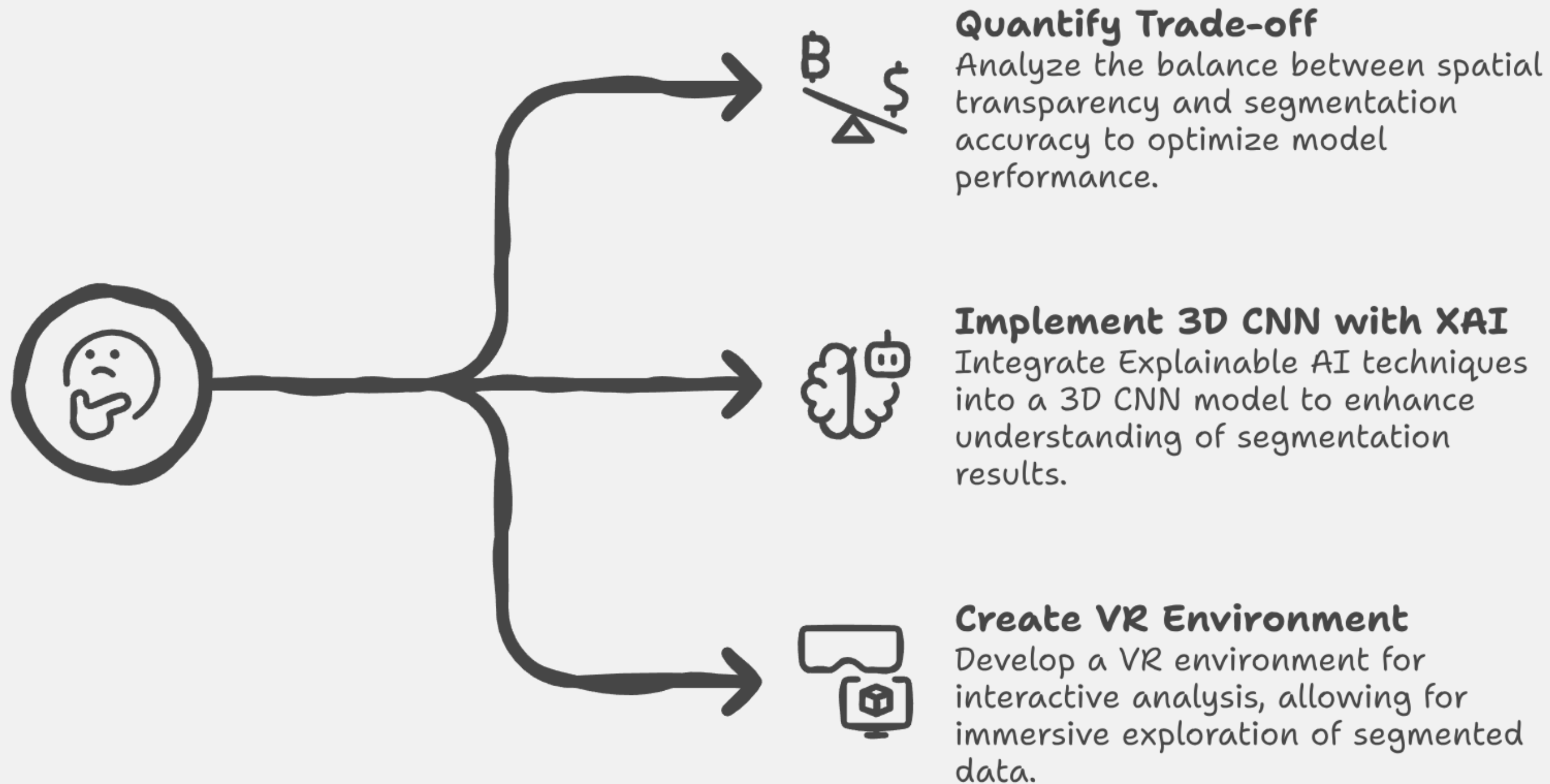
## • THE PROBLEM LANDSCAPE

- 3D MODELS ACHIEVE HIGH ACCURACY BUT REMAIN OPAQUE
- IN HIGH-STAKES NEUROSURGERY, UNCERTAINTY IS NOT AN OPTION
- CLINICIANS NEED THE **“WHY”** BEHIND THE CLASSIFICATION

## • PROJECT AIM

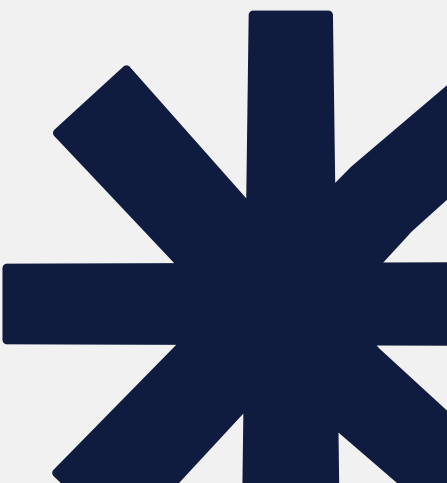
- ILLUMINATE THIS BLACK BOX NATURE USING EXPLAINABLE AI (XAI)
  - BRIDGE THE GAP BETWEEN THE ALGORITHMIC ACCURACY AND CLINICAL TRUST

# The path to Hybrid Explainability



# Literature Review Achievements

1. **3D volumetric superiority:** confirmed that 3D CNNs outperform 2D models by capturing spatial context along the Z-axis (depth).
2. Analyzed the success of the 3D CNNs , specifically the use of the **Dice coefficient** to handle class imbalance( small tumor vs large background) .
3. **The Fidelity Gap:** significant discrepancies between prediction and ground truth, hidden by model opacity.
4. **Cognitive Friction:** Increased error rates caused by analyzing 3D pathology on 2D screens.

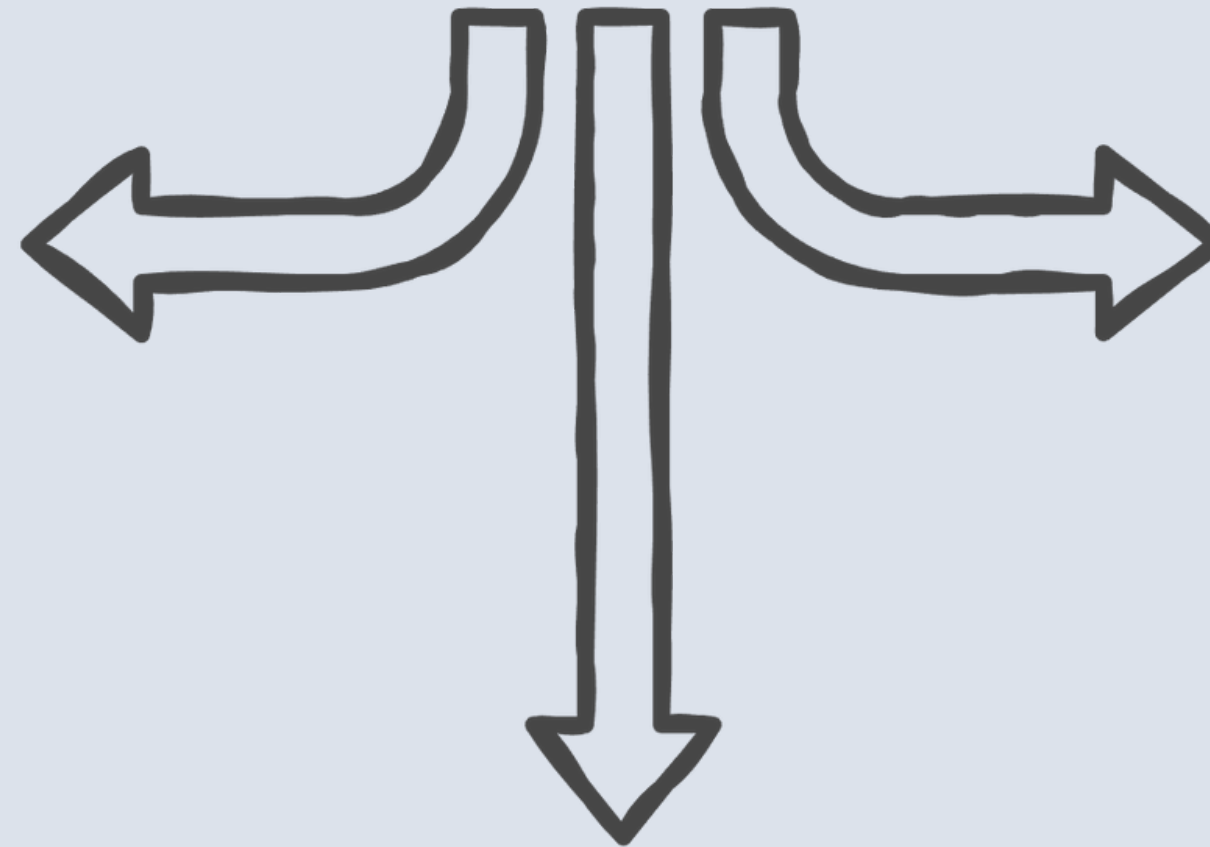




# Evaluation Metrics

Which evaluation metric should be used for 3D CNN segmentation?

**Dice Similarity Coefficient**  
Measures overlap between ground truth and predicted area, suitable for general segmentation accuracy.



**Intersection over Union (IoU)**  
Secondary measure of spatial overlap, useful for confirming accuracy.

**Hausdorff Distance (HD95)**  
Evaluates boundary distance, critical for surgical precision.



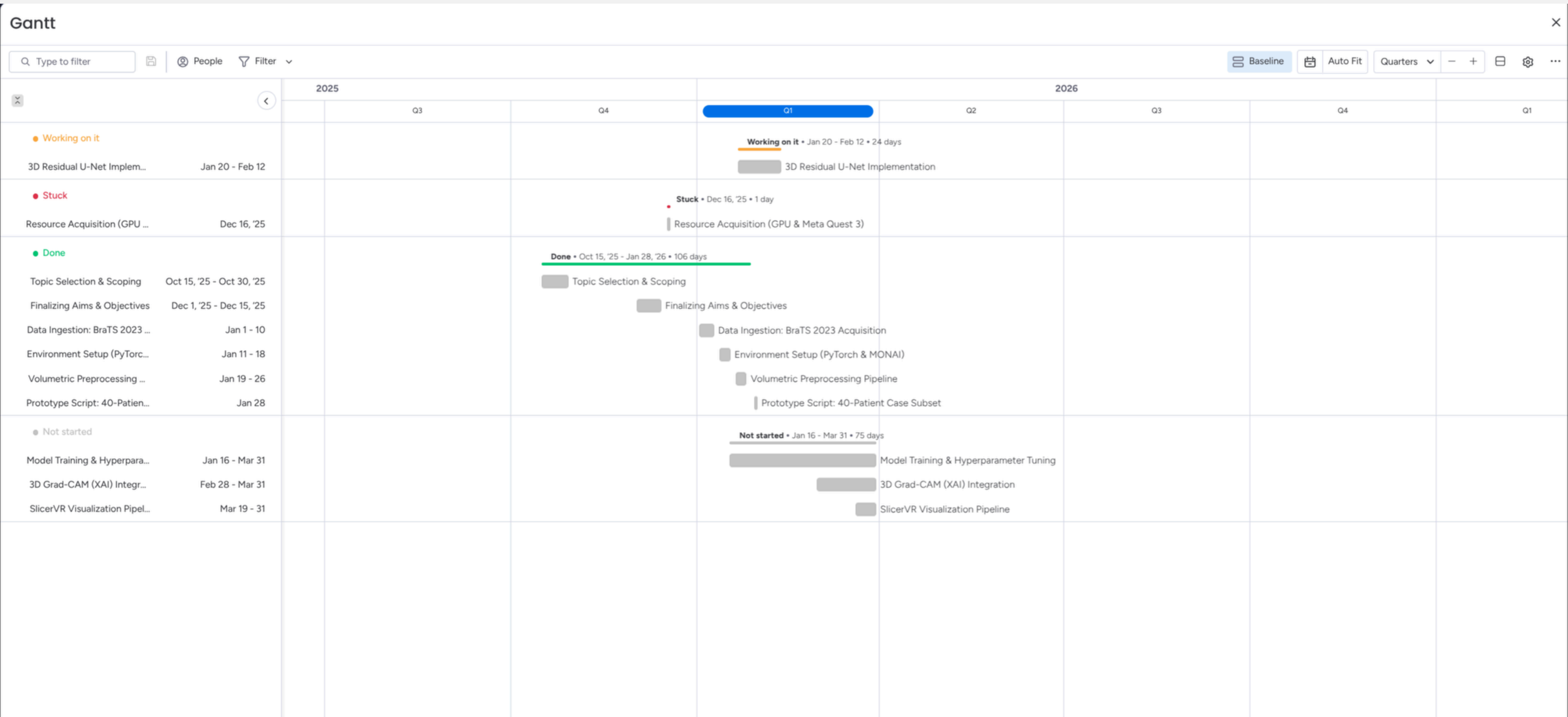
# Methodology & Implementation

## Current Accomplishment

- pipeline Development and project implementation planning.
- Phase 01: Data Engineering & preprocessing
  - Volumetric **Augmentation** implemented to expand dataset diversity.
  - Data cleaning and formatting is finalised.
- Working on Constructing 3D CNN model to test on preprocessed data.



# Gantt Chart



# Thank You

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## Q & A

